

CS111 Reference Document

String Methods

Method Signature	Description
int length()	Provides the number of characters in the string
char charAt(int index)	Returns the character at an index
int indexOf(String str)	Returns the index of the given string
String substring(int startIndex, int endIndex)	Returns a substring between the starting and ending index, exclusive of end
boolean equals(String str)	Returns true if both strings are the same
boolean equalsIgnoreCase(String str)	Returns true if both strings are the same, case-insensitive
int compareTo(String str)	Compares two strings to each other, returning the difference in ASCII values
int compareToIgnoreCase(String str)	Compares two strings to each other, returning the difference in ASCII values, case-insensitive
String[] split(String regex)	Splits the string by a separator
String strip()	Removes whitespace from both ends
char[] toCharArray()	Converts the string into an array of characters

StringBuilder Methods

Method Signature	Description
StringBuilder append(String s)	Adds the string to the StringBuilder
StringBuilder delete(int startIndex, int endIndex)	Deletes characters between the indexes, exclusive of end index
StringBuilder deleteCharAt(int index)	Deletes the character at the given index
String toString()	Converts the StringBuilder into the String version

ArrayList/LinkedList Methods

Method Signature	Description
E get(int index)	Retrieves the value at the given index
boolean add(E element)	Adds the element to the List

Method Signature	Description
void add(int index, E element)	Adds the element at the index to the List
boolean contains(E element)	Checks if the element is in the list
int indexOf(E element)	Returns the index of an element
E remove(int index)	Removes the value at an index
E set(int index, E element)	Sets the value at an index to the element
int size()	Returns the number of elements in the list

HashSet Methods

Method Signature	Description
boolean add(E element)	Adds the element to the Set
boolean contains(E element)	Checks if the element is in the set
boolean remove(E element)	Removes the element
int size()	Returns the number of elements in the set

HashMap Methods

Method Signature	Description
boolean containsKey(Object key)	Returns whether the key is in the map
V get(Object key)	Returns the value at the key, null if not found
boolean isEmpty()	Returns whether the map is empty
int size()	Returns how many elements are in the map
V put(K key, V value)	Puts the key-value pair in the map, replaces if the value already exists
V putIfAbsent(K key, V value)	Puts the key-value pair in the map if-and-only-if not already present
V getOrDefault(K key, Object defaultValue)	Gets the value at a key or a default value if not present